

SAADI Ahmed Abderrahim

+213 793915792 | ma_saadi@esi.dz | linkedin.com/in/ahmed-saadi-75116a335/ | github.com/ahmed6xx

EDUCATION

École Nationale Supérieure d'Informatique (ESI Algiers)

Algiers, Algeria

Engineering Degree in Computer Science – Intelligent Systems & Data (SID)

Sept. 2022 – June 2027 (Expected)

- Relevant Coursework: Machine Learning, Advanced Machine Learning (Federated Learning), Optimization Techniques (Metaheuristics), High-Performance Computing (GPU/CUDA)

PROJECTS

Multi-Objective Hardware-Aware Neural Architecture Search | *Python, NSGA-II*

- Reviewed 20+ papers on evolutionary and surrogate-based HW-NAS (NSGA-Net, SONATA), identifying blind mutation and absent local search as key gaps
- Addressed these gaps with CB-Adaptive VNS: a contextual bandit for adaptive mutation combined with Variable Neighbourhood Search for local refinement
- Benchmarked on NAS-Bench-201 across 6 hardware targets, outperforming baselines on constrained accelerators (Eyeriss, FPGA)

Transfer Learning Strategies for Medical Imaging | *Python, PyTorch, ResNet-50*

- Compared 5 transfer strategies on two MedMNIST datasets with contrasting class imbalance ($IR = 2.7$ vs. 58.7)
- Showed feature-space quality outweighs loss design: label-free SimCLR pretraining beat all class-balancing techniques

Green-HEAD: Energy-Aware Graph Coloring | *Python*

- Extended the HEAD memetic algorithm with SIMD encoding and adaptive k -compression for energy-efficient search
- Achieved 88.9% success rate (vs. 11.1% baseline) while cutting energy consumption by 75.9% on DIMACS benchmarks

CUDA-Accelerated Neural Network Training | *CUDA, C++*

- Diagnosed memory-transfer overhead as the main GPU training bottleneck ($\sim 80\%$ of runtime)
- Redesigned the pipeline with shared memory tiling and kernel fusion, achieving up to $36\times$ speedup

EXPERIENCE

LCSI Laboratory, ESI

Algiers, Algeria

Research Intern — Data Storytelling Dashboard

Sept. 2025

- Built an automated Exploratory Data Analysis pipeline (pandas, NumPy) computing missing-value ratios, feature correlations, and distributions, exported as structured JSON
- Designed an interactive dashboard (Plotly, Panel) translating analytical output into visual summaries for non-specialist users
- Integrated an LLM-based narrative layer (LLaMA-3.3-8B) to generate natural-language explanations of dataset trends and anomalies

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL

AI/ML: PyTorch, scikit-learn, NumPy, pandas, Matplotlib

Systems & Tools: CUDA, Git, Linux (Arch), LaTeX, VS Code